

# Wasi M.F. Naqvi

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## Education

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| 2026–   | <b>PhD Physics, McGill University</b><br>Supervisor: Nicolas B. Cowan<br>Thesis: A Patchwork of Worlds: Artificial Intelligence for Population-Level Studies of Exoplanet Atmospheres                                                                    | GPA: 4.00/4.00 |
| 2025–26 | <b>M.Sc. Physics, McGill University</b><br>Supervisor: Nicolas B. Cowan<br>Thesis: HERMES: HiERarchical Modelling for Exoplanet Science                                                                                                                  | GPA: 4.00/4.00 |
| 2021–25 | <b>B.Sc. Physics (Honours), The University of British Columbia</b><br>Minor in Data Science and Statistics<br>Thesis: Simulating In-Orbit Performance for the CASTOR Mission<br>Supervisor: Tyrone Woods<br>Distinctions: Dean's List, Best Thesis Award | GPA: 4.00/4.33 |

## Publications

**Naqvi, W. M. F.** and Cowan, N. B. *HERMES: HiERarchical Modelling for Exoplanet Science*. Royal Astronomical Society, Techniques and Instruments (2026, Submitted, under review.). [Available on arxiv](#)

Côté et al. (including **Naqvi, W.**). *The CASTOR Mission*. Special Issue, Journal of Astronomical Telescopes, Instruments, and Systems, SPIE. (2025)

## Awards & Honours

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| 2025–   | Kharusi Family International Science Fellowship, McGill University | \$15,000 CAD  |
| 2021–25 | Karen McKellin International Leader of Tomorrow Award, UBC         | \$250,000 CAD |
| 2025    | McGill Graduate Excellence Award                                   | \$5000 CAD    |

## Contributed Talks & Posters

 Contributed Talk    Poster

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|  Too Much Universe, Not Enough Grad Students: Deep Learning in Astrophysics<br>Trottier Space Institute, McGill University | Aug 2026<br>Montréal, Canada |
|  HERMES: HiERarchical Modelling for Exoplanet Science<br>CASCA Annual General Meeting.                                     | Jun 2026<br>Montréal, Canada |
|  Comparative climatology of exoplanets using Hierarchical Modelling,<br>Trottier Institute for Research on Exoplanets      | May 2026<br>Montréal, Canada |
|  <b>HERMES: HiERarchical Modelling for Exoplanet Science</b><br>Ariel Open Conference '26                                  | Mar 2026<br>ESA Harwell, UK  |

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| ♻ BayesBall: Bayesian Modelling for sport analytics,<br>Trottier Space Institute Lunch Talks | Nov 2025<br>Montréal, Canada |
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| ♻ A detector simulation pipeline for the CASTOR mission ( <b>Invited</b> )<br>The CASTOR Mission Consortium, NRC Herzberg Astronomy & Astrophysics | Jun 2026<br>Victoria, Canada |
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## Research & Work Experience

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| <b>Trottier Institute for Research on Exoplanets, McGill University</b><br><i>Summer Researcher</i><br>Hierarchical Bayesian Modelling of planetary targets to probe population-level trends for the Ariel Space Mission. | Montréal, QC<br>May 2025–Present |
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| <b>NRC Herzberg Astronomy &amp; Astrophysics</b><br><i>Junior Adaptive Optics Research Scientist</i><br>Implementing a convolutional neural network on an Adaptive Optics test bench for the 1.2 m REVOLT Telescope at the Dominion Astrophysical Observatory. Running machine-learning simulations alongside lab experiments under Dr. Maaïke van Kooten. | Victoria, BC<br>Jan 2025–Present |
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| <b>TRIUMF</b><br><i>Universal Polarization Research Assistant</i><br>Devising a Faraday Rotation method to measure polarization of optically-pumped Rb vapour at ISAC-II. Writing numerical simulations for polarizer experiments and developing a universal laser-polarization method for isotopes with unknown atomic transitions. | Vancouver, BC<br>Sep 2024–Present |
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| <b>NRC Herzberg Astronomy &amp; Astrophysics</b><br><i>Junior Research Scientist</i><br>Designed Python software pipelines for imaging-sensor simulations and detector modelling for the CASTOR mission (Canadian Space Agency / NRC). Built with Dask (parallel computing) and ESA Pyxel (detector physics). | Victoria, BC<br>May–Aug 2024 |
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| <b>COCANA Lab, UBC Okanagan</b><br><i>Undergraduate Research Assistant</i><br>Optimization research for a commercial road-building project (SoftTree Inc.) under Dr. Yves Lucet. Used YALMIP solver; translated codebases between MATLAB and Python with unit-test verification. | Kelowna, BC<br>May–Aug 2023 |
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| <b>CAN-SBX Stratospheric Balloon Design Team</b><br><i>Physics &amp; Data Science Lead</i><br>Led physics instrumentation and data analysis. Developed efficient MicroPython / C++ firmware for flight micro-controllers, designed ML models for atmospheric prediction, and assisted with dual-polarising antenna calibration and circuit analysis. | Kelowna, BC<br>Sep 2023–May 2024 |
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## Teaching Assistantships

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| <b>PHYS 180</b> Space, Time and Matter | Sep-Dec 2025 |
| <b>PHYS 241</b> Signal Processing      | Jan-Apr 2025 |

## Technical Competitions

Primary role across competitions: building, packaging, and benchmarking Bayesian models, U-Nets, Transformers, and CNNs for scientific data challenges.

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|-------------------------------------------------------------|----------|
| ExoHack: Ariel Machine Learning Hackathon ( <b>Winner</b> ) | Mar 2026 |
| NeurIPS Ariel Data Challenge                                | Jun 2026 |
| Rocky Worlds DDT Challenge                                  | May 2026 |

## Community Engagement & Outreach

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| Founding Member & Organizing Committee, Montréal AstroStats Day | 2026– |
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| <a href="#">Author, Astrobites</a>                                                         | 2025–     |
| Mentor & Teaching Assistant, Small Private Online Courses, Scientists for Displaced People | 2026–     |
| iREx EDI Committee Member                                                                  | 2025–     |
| McGill Hackathon Mentor                                                                    | 2026–     |
| Astro on Tap Volunteer                                                                     | 2026–     |
| Volunteer, McGill Physics “World of STEM” high-school astronomy sessions                   | 2025–     |
| Coordinator, graduate student lunches with TSI seminar speaker                             | 2025–     |
| Facilitator, McGill “From Planets to Particles” family science outreach fair               | 2025–     |
| Organizing & Administration Volunteer, Exoclines 2025 Conference                           | 2025–     |
| Science Outreach, Trottier Institute for Research on Exoplanets                            | 2025–     |
| UBCO Physics Student Club Treasurer                                                        | 2021–2025 |
| Exercise is Medicine Coordinator                                                           | 2023–2025 |
| Volunteer Soccer Coach, Royal Oak, Victoria                                                | 2024–2025 |
| Mentor, CERN BeamLine For Schools (BL4S), Cedar College, Karachi                           | 2021–     |
| Science Outreach, Let’s Talk Science Okanagan                                              | 2023–2024 |
| Volunteer, Okanagan Food Bank                                                              | 2022–2024 |
| Volunteer, Student Union Okanagan, UBC                                                     | 2022–2023 |

## Skills

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**Programming:** Python, C, C++, Julia, Fortran, JavaScript, R, SQL, Shell/Bash

**Markup & Typesetting:** LaTeX, HTML/CSS, Markdown

**Python Ecosystem:** PyTorch, JAX, NumPyro, PyMC, Emcee, NumPy, SciPy, Astropy, Scikit-learn, Dask, Matplotlib, Pandas, ArviZ, Corner, Dynesty

**Statistical & ML Methods:** Hierarchical Bayesian Modelling, Gaussian Processes, MCMC Sampling, Nested Sampling, Simulation-Based Inference, PCA, CNNs, U-Nets, Transformers, Variational Inference, Model Comparison and Benchmarking

**N-Body Codes:** REBOUND

**Computing Infrastructure:** CANFAR, DRAC Supercomputing Clusters, Slurm, MPI, Git/GitHub, Docker, Linux

**Other Software:** Microsoft Excel, Google Sheets

## Languages

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**Fluent:** English, Urdu, Farsi (Persian)

**Conversational:** Hindi

**Basic:** French, Arabic

## Relevant Coursework

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Radiative Processes in Astrophysics, Statistical Learning, Computational Astrophysics, Machine Learning Theory, Probability Theory, Stochastic Processes, Exoplanets and Brown Dwarfs.